Dominik Arih

.NET-Based Full-Stack Developer

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PROFESSIONAL SUMMARY

Results-driven Full-Stack Software Engineer with 4+ years of specialized expertise in .NET (C#) and modern web technologies. Proven track record of rapidly mastering new technologies and delivering high-impact solutions across diverse domains including enterprise applications, microservices architecture, desktop automation systems, and energy management platforms. Recognized for speed, precision, and efficiency in translating complex requirements into scalable, maintainable code. Adept at seamlessly transitioning between technology stacks while consistently maintaining high code quality and meeting aggressive deadlines. Demonstrated ability to lead technical initiatives, achieve superior code coverage, and collaborate effectively with cross-functional teams and government partners on mission-critical projects.

CORE COMPETENCIES

- **Advanced .NET Development: C#, ASP.NET Core, Entity Framework Core, LINQ, Dependency Injection**
- **Modern Frontend Frameworks: React, TypeScript, Vue.js, Progressive Web Apps (PWA)**
- **Backend Technologies: Python (Flask), RESTful APIs, Microservices Architecture**
- **Desktop Application Development: Avalonia, WPF, Signal-Based Communication Systems**
- **Database Management: PostgreSQL, MySQL, Advanced Query Optimization, Schema Design**
- **DevOps & Tools: Git, Docker, CI/CD Pipelines, Agile/Scrum Methodologies**
- **Additional Expertise: Computer Vision, AI/Robotics Integration, Unity, Unreal Engine**
- **Rapid Technology Adoption: Proven ability to quickly master and effectively deploy new frameworks, languages, and architectural patterns**

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer

UPRISE D.O.O. | Zagreb, Croatia | September 2022 – Present

Lead full-stack development initiatives across 10+ enterprise-grade applications using .NET (C#), React (TypeScript/JavaScript), Vue.js, and Python. Collaborate directly with government ministries and private sector clients to architect, implement, and deploy scalable solutions. Consistently achieve the high code coverage while meeting critical deadlines for mission-critical systems.

Key Projects & Achievements:

STORM - Energy Management System (December 2025 – Present)

- Developing full-stack energy management platform with Python Flask backend and React/TypeScript frontend
- Working on RESTful APIs for real-time energy monitoring, data visualization, and predictive analytics
- Working on responsive data dashboards with interactive charts enabling actionable insights for energy optimization

CRiSP - Insolvency Data Collection System (February 2024 – December 2024)

- Partnered with Ministry of Justice and Public Administration on nationwide insolvency monitoring system
- Engineered comprehensive report generation system with hierarchical tree visualization for complex temporal data analysis
- Developed robust export functionality supporting multiple formats (PDF, Excel, CSV) with customizable filtering

eSavjetnik - Agricultural Advisory Platform Enhancement (April 2024 – February 2025)

- Modernized government agricultural information system using microservices architecture with .NET and React
- Integrated National Identification and Authentication System (NIAS) ensuring secure multi-ministry data access
- Implemented real-time advisory features serving thousands of agricultural stakeholders

Abysalto Vivax - HVAC Service Management PWA (September 2025 – January 2026)

- Built Progressive Web Application centralizing installation and maintenance workflows for HVAC technicians

eETIKA - Civil Servant Ethics Digitalization (August 2023 – August 2024)

- Collaborated with Ministry of Justice and Public Administration on national ethics compliance platform
- Created dynamic reporting system with interactive graphs and temporal analytics, supporting PDF/Excel export

MINRAD-ISU - Social Services Cost Management System (May 2023 – June 2024)

- Developed pricing calculation engine for Ministry of Labour, Pension System, Family and Social Policy
- Architected comprehensive reporting framework with custom export capabilities

VTR - Foreign Trade Analytics Platform (November 2023 – June 2024)

- Partnered with Ministry of Agriculture, Forestry and Fisheries on trade monitoring system
- Built interactive geospatial visualization system displaying trade metrics by country with drill-down analytics

JAO - LTFBA Simulation Platform (June 2023 – August 2023)

- Engineered energy capacity allocation optimization module using Flow-Based methodology for European market
- Implemented XSD schema validation system for CFG parameters and FB domain submissions
- Developed version control and progress tracking features ensuring data integrity across complex workflows

Additional Key Contributions:

- **RPIS** - Developed reservation timetable system for world's first inland navigation community platform (Apr 2023 – Sep 2023)
- **RIC (ROSC Input Converter)** - Built data transformation tool for European transmission network congestion analysis (Sep 2023 – Nov 2023)
- **PINS & Aconto** - Created cost calculation systems and financial analytics modules for SME business management

Software Engineer & Researcher

BYTE MOTION | Västerås, Sweden (Remote) | April 2024 – October 2025

Specialized in artificial intelligence and computer vision applications for industrial robotics. Contributed to Optimus—an integrated platform for Industrial Automation, AI, and IoT—collaborating with international research teams on cutting-edge automation solutions.

- Developed computer vision modules using Python and C++ for real-time object recognition and automated decision-making
- Implemented stereoscopic processing algorithms for robotic perception systems with sub-millimeter accuracy
- Collaborated with international universities on AI-driven robotics research, advancing automation capabilities

Desktop Application Developer

DOK-ING Remote Controller Project | March 2025 – September 2025

Developed mission-critical remote control system for military vehicle operations using Avalonia framework and C#.

- Architected signal-based communication layer integrating aeronautical controller hardware with vehicle command systems
- Ensured system reliability through comprehensive error handling and failsafe mechanisms for defense applications

EDUCATION

Master of Science in Computer Science | University of Zagreb, Faculty of Electrical Engineering and Computing

September 2022 – September 2024

- **Thesis:** Digital Twin of a Robotic System for Cutting Objects with User Correction through Virtual Reality
- **Relevant Coursework:** Machine Learning, Computer Vision, Interactive Simulation Systems, Distributed Systems, IoT, Cloud Computing

Bachelor of Science in Computer Science | University of Zagreb, Faculty of Electrical Engineering and Computing

September 2019 – September 2022

- **Thesis:** Application of Pathfinding Algorithms in Non-Player Characters Behavior in a Video Game
- **Relevant Coursework:** Algorithms & Data Structures, Software Engineering, Object-Oriented Programming, Artificial Intelligence

PUBLICATIONS & RECOGNITION

- **Published Researcher** – IEEE SoftCOM 2024 Conference: "Determination of the Impact of Video Encoding Parameters on QoE Using Passive Viewing Tests" (Arih et al., 2024)
- **Rector's Award** – University of Zagreb, 2021/2022: Team scientific work in technical sciences for educational game "The Witch's Secret" (cryptography education)

TECHNICAL SKILLS

Languages & Frameworks: C#, .NET (ASP.NET Core, Entity Framework Core), Python (Flask), TypeScript, JavaScript (React, Vue.js), Java, C++

Databases: PostgreSQL, MySQL, SQL Server, Entity Framework Core, LINQ

DevOps & Tools: Git, Docker, CI/CD Pipelines, Agile/Scrum, Azure DevOps

Desktop Development: Avalonia, WPF, Windows Forms

Specialized: Computer Vision (OpenCV), AI/ML (TensorFlow, PyTorch), Unity, Unreal Engine, ROS

Methodologies: Microservices Architecture, RESTful API Design, Test-Driven Development (TDD), Clean Code Principles

LANGUAGES

Croatian: Native

English: Fluent (Professional working proficiency)

German: Basic